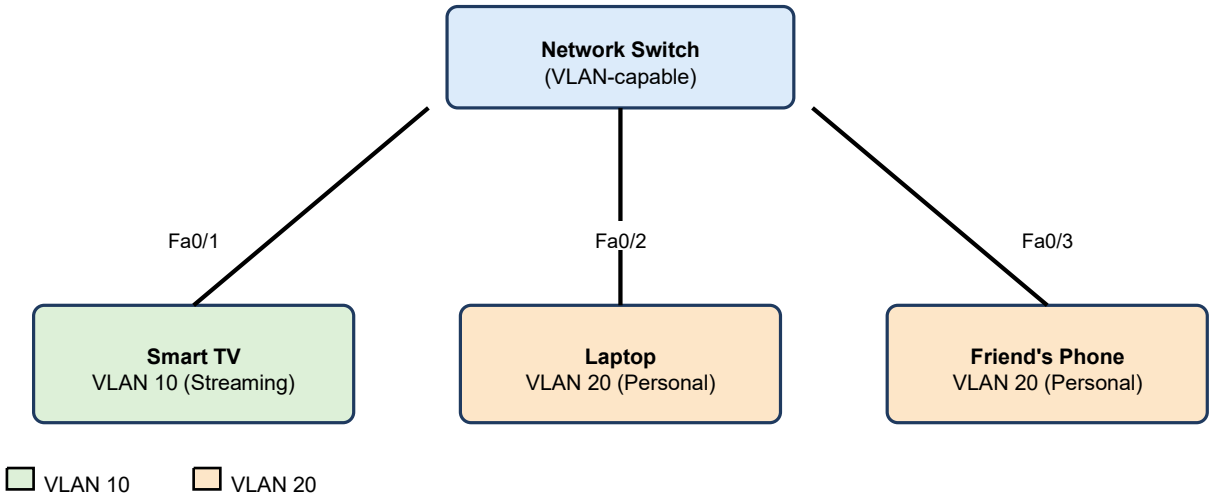
SWITCHING AND VLANS

TASK-1 Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

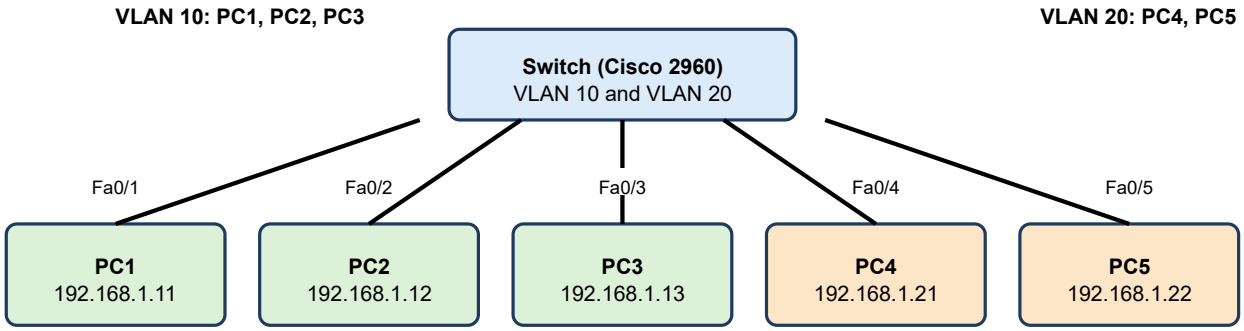


Explanation

- All three devices connect to one physical switch, but the switch separates them into two logical networks (VLANs).
- The Smart TV is in **VLAN 10** (streaming); the laptop and the friend's phone are in **VLAN 20** (personal devices).
- Devices in the same VLAN can talk to each other. Devices in different VLANs cannot, unless a router (or Layer 3 switch) connects them.

SWITCHING AND VLANS

TASK-2 Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.Hint Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.



Tool used: Cisco Packet Tracer. All PCs use subnet mask 255.255.255.0 and are in the same IP network (192.168.1.0/24), so any ping failure is caused by VLAN separation only.

Switch configuration

Verification: show vlan brief

```
Switch# show vlan brief
VLAN Name       Status    Ports
---- ---------- --------- -------------------------------------1    default    active    Fa0/6,
Fa0/7, Fa0/8, Fa0/9, Fa0/10 ...
10   Group-A    active    Fa0/1, Fa0/2, Fa0/3
20   Group-B    active    Fa0/4, Fa0/5
```

Ping tests

```
C:\> ping 192.168.1.21          (PC1 in VLAN 10 to PC4 in VLAN 20) Pinging 192.168.1.21 with 32
bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.1.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

SWITCHING AND VLANS

TASK-3 In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

Scenario: PC3 (192.168.1.13) is moved from VLAN 10 to VLAN 20 by changing its switch port (Fa0/3).

BEFORE the change

```
Switch# show vlan brief
10   Group-A    active    Fa0/1, Fa0/2, Fa0/3
20   Group-B    active    Fa0/4, Fa0/5

C:\> ping 192.168.1.11      (PC3 to PC1)   -> Reply received (0% loss)
C:\> ping 192.168.1.21      (PC3 to PC4)   -> Request timed out (100% loss)
```

Configuration commands used for the move

```
Switch# configure terminal
Switch(config)# interface fastEthernet 0/3
Switch(config-if)# switchport access vlan 20
Switch(config-if)# end
Switch# write memory
```

AFTER the change

```
Switch# show vlan brief
10   Group-A    active    Fa0/1, Fa0/2
20   Group-B    active    Fa0/3, Fa0/4, Fa0/5

C:\> ping 192.168.1.21      (PC3 to PC4)   -> Reply received (0% loss)
C:\> ping 192.168.1.11      (PC3 to PC1)   -> Request timed out (100% loss)
```

Test	Before (PC3 in VLAN 10)	After (PC3 in VLAN 20)
PC3 to PC1 (VLAN 10)	Success	Fail
PC3 to PC4 (VLAN 20)	Fail	Success

SWITCHING AND VLANS

TASK-4 Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3)

Simulator used: Cisco Packet Tracer (Cisco 2960 switch). The **interface range** command applies one set of commands to ports Fa0/1 to Fa0/10, so no command is repeated 10 times.

Command list

```
enable configure terminal vlan 30
name VLAN30  exit interface range
fastEthernet 0/1 - 10  switchport
mode access  switchport access vlan
30  exit end write memory show vlan
brief
```

What each part does

Command	Purpose
vlan 30 / name VLAN30	Creates VLAN 30 in the switch database and gives it a name.
interface range fastEthernet 0/1 - 10	Selects 10 ports (Fa0/1 to Fa0/10) at once; this is the loop/range that avoids repeating the command 10 times.
switchport mode access	Makes the selected ports access ports, which carry traffic for one VLAN only (used for end devices).
switchport access vlan 30	Assigns all selected ports to VLAN 30.
write memory	Saves the configuration so it is not lost after a reboot.
show vlan brief	Verifies that Fa0/1 to Fa0/10 now appear under VLAN 30.

Expected verification output

```
VLAN Name       Status    Ports
---- ---------- --------- ----------------------------------------
30   VLAN30     active    Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5,
                          Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10
```